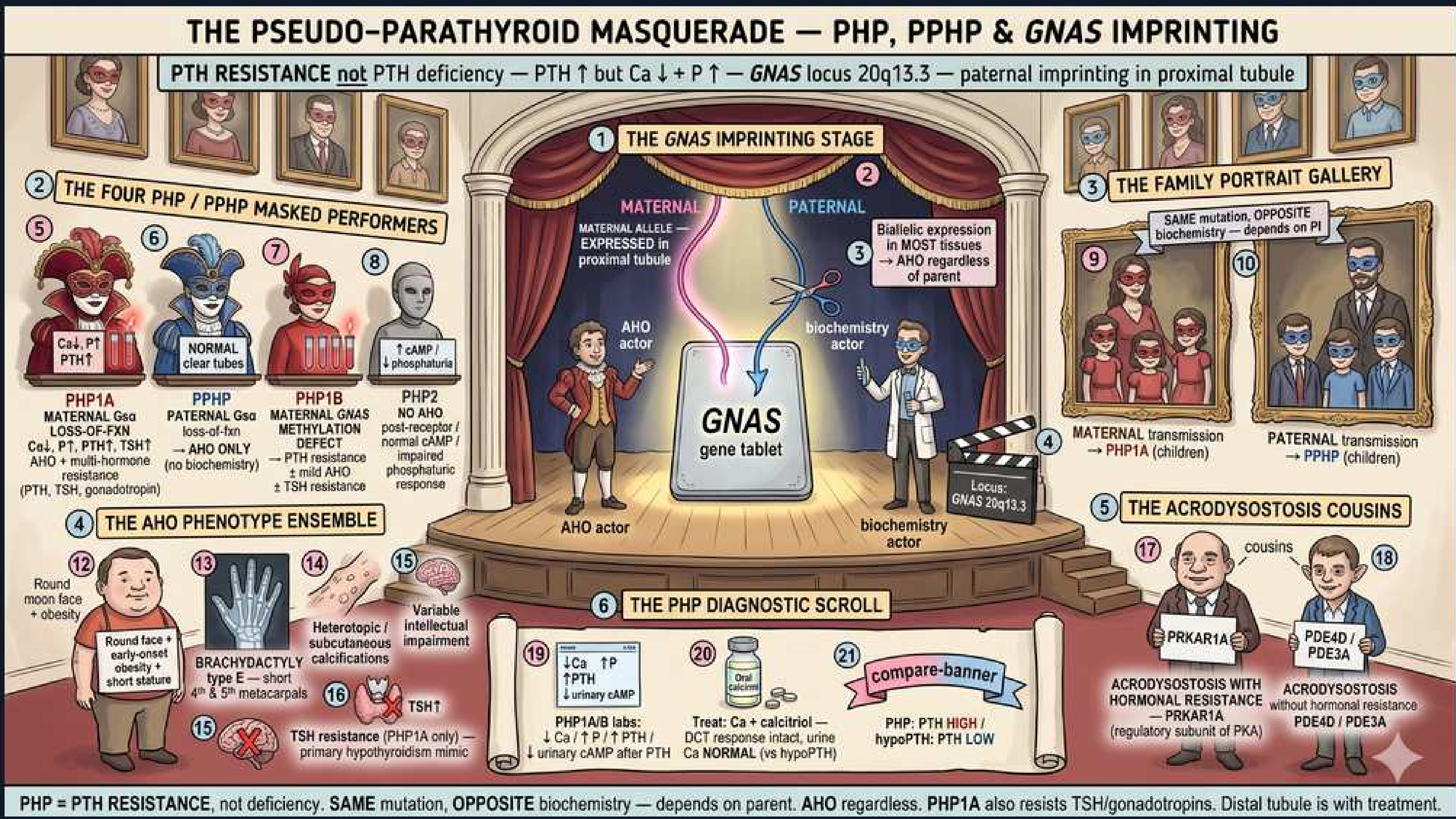
Pseudohypoparathyroidism — PHP, PPHP & GNAS Imprinting



1. PTH resistance, not deficiency

WHAT YOU SEE

Banner: PTH resistance not PTH deficiency

WHAT IT ENCODES

Pseudohypoparathyroidism is end-organ PTH resistance: PTH is HIGH but calcium is LOW and phosphate is HIGH. Contrast true hypoparathyroidism where PTH is low. The lesion is in the GNAS-encoded Gs-alpha signaling pathway in the proximal tubule, so PTH cannot raise cAMP.

BOARD TRAP

Wrong answer: hypoparathyroidism (low PTH). Discriminator: in PHP the PTH is markedly ELEVATED despite hypocalcemia.

2. GNAS locus 20q13.3

WHAT YOU SEE

GNAS gene tablet center stage

WHAT IT ENCODES

GNAS at 20q13.3 encodes Gs-alpha. It is an imprinted gene: the maternal allele is preferentially expressed in proximal renal tubule. Maternal loss-of-function mutations therefore cause renal PTH resistance (PHP1A); paternal mutations do not.

BOARD TRAP

Wrong answer: a defect in the PTH receptor itself. Discriminator: the defect is downstream in Gs-alpha (GNAS), not the PTHR1 receptor.

3. PHP1A — maternal, AHO + multi-resistance

WHAT YOU SEE

PHP1A masked performer, Gs-alpha loss

WHAT IT ENCODES

PHP1A: maternal GNAS loss-of-function. Features AHO phenotype PLUS multi-hormone resistance (PTH, TSH, gonadotropins) because Gs-alpha is reduced in many tissues. Low calcium, high phosphate, high PTH, blunted urinary cAMP after exogenous PTH (Ellsworth-Howard test).

BOARD TRAP

Wrong answer: PHP1B. Discriminator: PHP1A has AHO morphology and TSH/gonadotropin resistance; PHP1B is renal-restricted without AHO.

4. PPHP — paternal, AHO only

WHAT YOU SEE

PPHP performer, normal calcium

WHAT IT ENCODES

Pseudopseudohypoparathyroidism: paternal GNAS loss-of-function. Same AHO physical features but NORMAL calcium/phosphate and NO hormone resistance, because maternal allele still functions in proximal tubule.

BOARD TRAP

Wrong answer: PHP1A. Discriminator: PPHP has AHO appearance but NORMAL labs and no resistance (paternal inheritance).

5. PHP1B — renal GNAS methylation

WHAT YOU SEE

PHP1B methylation, no AHO

WHAT IT ENCODES

PHP1B: imprinting/methylation defect of maternal GNAS limited to proximal tubule. PTH (and sometimes mild TSH) resistance WITHOUT AHO features and without generalized Gs-alpha deficiency. Labs mimic PHP1A.

BOARD TRAP

Wrong answer: PHP1A. Discriminator: PHP1B lacks AHO skeletal features and the multi-tissue Gs-alpha reduction.

6. PHP2 — normal cAMP, no phosphaturia

WHAT YOU SEE

PHP2, cAMP normal but no phosphaturic response

WHAT IT ENCODES

PHP2: PTH normally raises urinary cAMP but the downstream phosphaturic response fails. Distinguishes from PHP1 (where cAMP generation itself is blunted). Rare; some cases linked to vitamin D deficiency mimicking PHP2.

BOARD TRAP

Wrong answer: PHP1A/1B. Discriminator: PHP2 has a NORMAL urinary cAMP rise after PTH but absent phosphaturia.

7. AHO phenotype (Albright osteodystrophy)

WHAT YOU SEE

Short stocky figure, round face

WHAT IT ENCODES

Albright hereditary osteodystrophy: short stature, round face, obesity, brachydactyly (short 4th/5th metacarpals), subcutaneous ossifications, and variable intellectual disability. Appears in both PHP1A and PPHP.

BOARD TRAP

Wrong answer: this proves PTH resistance. Discriminator: AHO appears in PPHP too, which has NORMAL calcium — phenotype alone does not equal resistance.

8. Brachydactyly — short 4th/5th metacarpal

WHAT YOU SEE

Hand X-ray with short metacarpals

WHAT IT ENCODES

Brachydactyly type E: shortened 4th and 5th metacarpals, classically demonstrated by a dimple over the knuckle when making a fist. A hallmark AHO skeletal sign on board imaging questions.

BOARD TRAP

Wrong answer: arachnodactyly/Marfan. Discriminator: AHO shows SHORT metacarpals (knuckle dimple), not long thin digits.

9. Subcutaneous ossifications

WHAT YOU SEE

Heterotopic calcification figure

WHAT IT ENCODES

Heterotopic/subcutaneous ossifications are a feature of AHO and can be the presenting sign in infancy. Reflect reduced Gs-alpha; seen in PHP1A and PPHP.

BOARD TRAP

Wrong answer: tumoral calcinosis from hyperphosphatemia alone. Discriminator: AHO ossifications are part of the genetic syndrome, present even in normocalcemic PPHP.

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10. TSH resistance → subclinical hypothyroidism

WHAT YOU SEE

TSH high banner

WHAT IT ENCODES

In PHP1A, Gs-alpha deficiency also blunts TSH receptor signaling, giving mild TSH elevation with normal/low T4. Gonadotropin resistance may cause hypogonadism. Multi-hormone resistance is the signature of PHP1A.

BOARD TRAP

Wrong answer: autoimmune Hashimoto thyroiditis. Discriminator: PHP1A TSH resistance is part of generalized Gs-alpha loss, antibody-negative.

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11. Labs: PTH HIGH, Ca LOW, PO4 HIGH

WHAT YOU SEE

PTH HIGH + hypoCa + hyperPO4 box

WHAT IT ENCODES

Diagnostic triad of PHP: elevated PTH, hypocalcemia, hyperphosphatemia. Same biochemistry as hypoparathyroidism EXCEPT PTH is high (resistance) not low. Confirm with genetics or Ellsworth-Howard test.

BOARD TRAP

Wrong answer: primary hyperparathyroidism (high PTH, high Ca). Discriminator: PHP has high PTH with LOW calcium, the opposite of primary HPT.

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12. Acrodysostosis cousins — PRKAR1A / PDE4D

WHAT YOU SEE

Acrodysostosis cousins, PRKAR1A/PDE4D

WHAT IT ENCODES

Acrodysostosis resembles AHO/PHP but is caused by PRKAR1A (PKA regulatory subunit) or PDE4D mutations. PRKAR1A type has hormone resistance; PDE4D type usually does not. Phenocopies in the cAMP/PKA pathway.

BOARD TRAP

Wrong answer: all AHO-like skeletal disease is GNAS. Discriminator: acrodysostosis is PRKAR1A/PDE4D, distinct from GNAS-based PHP.

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13. Same mutation, opposite biochemistry

WHAT YOU SEE

Family portrait gallery, parent-of-origin

WHAT IT ENCODES

Parent-of-origin imprinting explains why the IDENTICAL GNAS mutation gives PHP1A when maternally inherited but PPHP when paternally inherited. Tests love this maternal-vs-paternal transmission distinction.

BOARD TRAP

Wrong answer: differing mutations cause the two phenotypes. Discriminator: SAME mutation; phenotype depends on parental origin (imprinting).
